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BASIC MATHEMATICS

FORM FOUR

Certificate of Secondary Education Examination (CSEE) 2021 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Four
Year	2021
Exam Board	NECTA
Exam	Certificate of Secondary Education Examination (CSEE)
Type	Answer key and worked solutions
Questions	14 compulsory questions; 38 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F4-2021.pdf

Question 14 mixes metre quantities and garment counts. Its solutions show the literal printed-unit interpretation and the alternative garment-count interpretation explicitly.

Section A — Worked solutions

1(a). Find the LCM of 15, 35 and 40.

Step 1: $15 = 3 \times 5$; $35 = 5 \times 7$; $40 = 2^3 \times 5$.

Step 2: Use the highest power of each prime: $2^3 \times 3 \times 5 \times 7 = 840$.

Final Answer: 840.

1(b). Approximate $(0.0695 \times 19812)/6.8125$ by first rounding each number to one significant figure.

Step 1: Round each input: $0.0695 \approx 0.07$; $19812 \approx 20,000$; $6.8125 \approx 7$.

Step 2: Compute $(0.07 \times 20,000)/7 = 1400/7 = 200$.

Final Answer: 200.

2(a). Express $3^{(2y-1)} + 2 \times 3^{(y-1)} = 1$ in terms of $P = 3^y$.

Step 1: $3^{(2y-1)} = (3^y)^2/3 = P^2/3$; $3^{(y-1)} = P/3$.

Step 2: Substitute: $P^2/3 + 2P/3 = 1$. Multiply by 3 and bring all terms to the left.

Final Answer: $P^2 + 2P - 3 = 0$.

2(b). Solve $\log_{10}(3y+2) - 1 = \log_{10}(y-4)$.

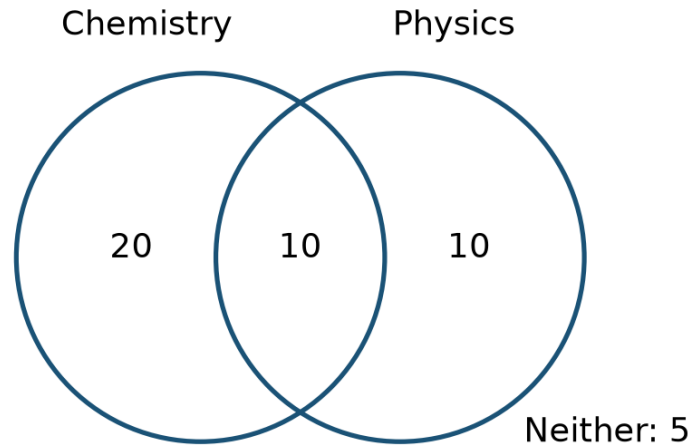
Step 1: The domain requires $y > 4$. Write $1 = \log_{10}10$ and combine: $\log_{10}[(3y+2)/10] = \log_{10}(y-4)$.

Step 2: Equate arguments: $3y+2 = 10y-40$, so $7y = 42$ and $y = 6$.

Step 3: Check $y-4 = 2 > 0$, and $(3y+2)/10 = 2$.

Final Answer: $y = 6$.

3(a). Of 45 students, 30 study Chemistry, 20 Physics and 5 neither. Draw a labelled Venn diagram.



Step 1: At least one subject = $45 - 5 = 40$. Overlap = $30 + 20 - 40 = 10$.

Step 2: Chemistry only = $30 - 10 = 20$; Physics only = $20 - 10 = 10$. Place the four counts in their appropriate regions.

Final Answer: Chemistry only = 20; both = 10; Physics only = 10; neither = 5.

3(b)(i). Find the number studying both subjects.

Step 1: The two subject totals count the overlap twice.

Step 2: Both = $30 + 20 - (45 - 5) = 10$.

Final Answer: 10 students.

3(b)(ii). Find the probability that a randomly selected student studies Chemistry only.

Step 1: Chemistry only = $30 - 10 = 20$.

Step 2: All 45 students are eligible for selection, so probability = $20/45 = 4/9$.

Final Answer: 4/9.

4(a). Road R_1 has equation $2x - 3y - 4 = 0$. Perpendicular road R_2 passes through $(4, -2)$. Find R_2 as $ax + by + c = 0$.

Step 1: R_1 has gradient $2/3$ after rearranging to $y = (2/3)x - 4/3$. A perpendicular gradient is $-3/2$.

Step 2: Through $(4, -2)$: $y + 2 = (-3/2)(x - 4)$. Multiply by 2 and rearrange to $3x + 2y - 8 = 0$.

Final Answer: $3x + 2y - 8 = 0$.

4(b). A boat moves 30 km/h south while river water flows 5 km/h east. Find the resultant velocity to two decimal places.

Step 1: Take east as positive x and north as positive y . Add vectors $(0, -30) + (5, 0) = (5, -30)$.

Step 2: Magnitude $= \sqrt{5^2 + 30^2} = \sqrt{925} \approx 30.4138$ km/h.

Step 3: Direction east of south satisfies $\tan \theta = 5/30$, so $\theta \approx 9.46^\circ$.

Final Answer: $(5, -30)$ km/h, with speed 30.41 km/h directed S 9.46° E (bearing 170.54°).

Study Tip: Velocity includes both magnitude and direction. The resultant speed alone is not a complete velocity description.

5(a). Similar rectangles have lengths 6 cm and 8 cm. The smaller area is 73.8 cm^2 . Find the larger area.

Step 1: Linear scale factor $= 8/6 = 4/3$; area scale factor $= (4/3)^2 = 16/9$.

Step 2: Larger area $= 73.8 \times 16/9 = 131.2 \text{ cm}^2$.

Final Answer: 131.2 cm^2 .

5(b). A regular polygon's exterior and interior angles are in ratio 2:4. Find its number of sides.

Step 1: Let exterior $= 2k$ and interior $= 4k$; together they sum to 180° . Then $6k = 180$, so $k = 30$.

Step 2: Exterior angle $= 60^\circ$; number of sides $= 360/60 = 6$.

Final Answer: 6 sides.

6(a). Cut 7.42 m from a 13.5 m string and divide the remainder into 32 cm pieces. How many pieces?

Step 1: Remaining length = $13.5 - 7.42 = 6.08 \text{ m} = 608 \text{ cm}$.

Step 2: $608/32 = 19$.

Final Answer: 19 pieces.

6(b). Supported mass varies directly with breadth b and inversely with length l . A 2 m wide, 15 m long beam supports 200 kg. Find mass for $b = 3 \text{ m}$, $l = 20 \text{ m}$.

Step 1: $M = kb/l$. Use $200 = k(2)/15$ to find $k = 1500$.

Step 2: For the new beam, $M = 1500 \times 3/20 = 225 \text{ kg}$.

Final Answer: 225 kg.

7(a)(i). Define trading account as used in Accounts.

Step 1: Find net sales and the cost of goods sold, accounting for stock and purchases.

Step 2: Subtract cost of goods sold from net sales; the resulting gross profit or loss belongs in the trading account.

Final Answer: An account used to calculate gross profit or gross loss from sales and cost of goods sold for a period.

7(a)(ii). Define profit and loss account.

Step 1: Start with gross profit or gross loss from the trading account.

Step 2: Add other income and deduct operating expenses to determine the net result for the period.

Final Answer: An account used to calculate net profit or net loss after considering operating expenses and other income.

7(a)(iii). Define balance sheet.

Step 1: Assets are resources owned; liabilities are obligations owed.

Step 2: Present them with capital/equity so that $\text{Assets} = \text{Liabilities} + \text{Equity}$.

Final Answer: A statement showing a business's assets, liabilities and owner's equity at a particular date.

7(a)(iv). Define cash account.

Step 1: Cash receipts are recorded on the debit side and cash payments on the credit side.

Step 2: Balance the account to determine the cash remaining.

Final Answer: An account recording cash received, cash paid and the resulting cash balance.

7(b). A car bought for Tsh 12,500,000 is sold at 20% loss. Find loss and selling price.

Step 1: Loss = $(20/100) \times 12,500,000 = 2,500,000$.

Step 2: Selling price = cost – loss = $12,500,000 - 2,500,000 = 10,000,000$.

Final Answer: Loss = Tsh 2,500,000; selling price = Tsh 10,000,000.

8(a). Six mango seedlings are planted in a row 7 m apart. Find the row length.

Step 1: Six seedlings create five gaps between consecutive seedlings.

Step 2: Length from first to last = $(6 - 1) \times 7 = 35$ m.

Final Answer: 35 m.

8(b). The fifth and sixth GP terms are 162 and 486. Find the common ratio and first term.

Step 1: Common ratio $r = T_6/T_5 = 486/162 = 3$.

Step 2: $T_5 = ar^4$, so $a = 162/3^4 = 162/81 = 2$.

Final Answer: $r = 3$; $a = 2$.

9(a). Show $\cos(90^\circ + \theta) = -\sin \theta$.

Step 1: Use $\cos(A + B) = \cos A \cos B - \sin A \sin B$.

Step 2: Substitute $A = 90^\circ$: $0 \times \cos \theta - 1 \times \sin \theta = -\sin \theta$.

Final Answer: $\cos(90^\circ + \theta) = -\sin \theta$.

9(b). $UV = 3$ cm, $UW = 5$ cm and their included angle is 60° . Find VW to two decimal places.

Step 1: Cosine rule: $VW^2 = 3^2 + 5^2 - 2(3)(5)\cos 60^\circ$.

Step 2: $VW^2 = 9 + 25 - 30(1/2) = 19$; $VW = \sqrt{19} \approx 4.3589$ cm, rounding to 4.36 cm.

Final Answer: 4.36 cm.

10(a). A trapezium has area $2x^2 - 8x + 6$ and parallel sides $2x + 3$ and $2x - 7$. Find its height.

Step 1: $A = (1/2)(\text{sum of parallel sides})h$ gives $2x^2 - 8x + 6 = (1/2)(4x - 4)h$.

Step 2: Factor: $2(x - 1)(x - 3) = 2(x - 1)h$, so $h = x - 3$.

Step 3: Both side lengths must be positive; $2x - 7 > 0$ requires $x > 7/2$, which also makes the height positive.

Final Answer: $h = x - 3$ units, for geometrically valid $x > 7/2$.

10(b). Two positive numbers differ by 7 and have product 30. Find them.

Step 1: Let the smaller be $s > 0$, so $s(s + 7) = 30$.

Step 2: $s^2 + 7s - 30 = (s + 10)(s - 3) = 0$. Reject -10 ; $s = 3$ and the larger is 10.

Final Answer: 10 and 3.

Section Summary: Use exact fractions, consistent units and the requested method. For vectors, give direction as well as speed; for similar figures, square the linear factor when comparing areas.

Section B — Worked solutions

11(a). Group the 40 marks into eight classes of width 8 starting at 32.

Step 1: Form non-overlapping classes using lower limits 32,40,48,56,64,72,80,88.

Step 2: Tally every raw mark exactly once. Check the frequencies $4 + 13 + 8 + 6 + 6 + 2 + 0 + 1 = 40$.

Final Answer: 32-39: 4; 40-47: 13; 48-55: 8; 56-63: 6; 64-71: 6; 72-79: 2; 80-87: 0; 88-95: 1. Total: 40.

11(b). Use the grouped distribution to find the mean with assumed mean 83.5.

Step 1: Class midpoints are 35.5, 43.5, 51.5, 59.5, 67.5, 75.5, 83.5, 91.5. Set $u = (\text{midpoint} - 83.5)/8$.

Step 2: The coded values are $-6, -5, -4, -3, -2, -1, 0, 1$. Multiply by frequencies: $-24, -65, -32, -18, -12, -2, 0, 1$; sum $\sum fu = -152$.

Step 3: Mean = $A + h(\sum fu / \sum f) = 83.5 + 8(-152/40) = 53.1$.

Final Answer: Grouped mean = 53.1 marks.

Study Tip: This is the grouped-data estimate, obtained by treating every mark in a class as its midpoint. It need not equal the exact mean of the ungrouped scores.

11(c). Find the difference between the mean and the median of the grouped distribution and comment.

Step 1: Cumulative frequencies are 4, 17, 25, 31, 37, 39, 39, 40. $N/2 = 20$ lies in class 48–55.

Step 2: Use median = $L + [(N/2 - \text{previous cumulative frequency}) / \text{class frequency}]h = 47.5 + [(20 - 17)/8] \times 8 = 50.5$.

Step 3: Difference = $53.1 - 50.5 = 2.6$; the average is slightly raised relative to the middle value by the higher marks.

Final Answer: Grouped median = 50.5; mean – median = 2.6 marks. The mean is slightly higher than the median.

12(a). For the cuboid, name the angles between AG and planes ABCD and BCFG.

Step 1: In the original labelling G lies vertically above B, so the projection of AG on base ABCD is AB. The line-plane angle is $\angle GAB$.

Step 2: A projects perpendicularly to B on the side plane BCFG, so the projection of AG there is GB. The angle at G is $\angle AGB$.

Final Answer: With ABCD: $\angle GAB$. With BCFG: $\angle AGB$.

12(b)(i). The block is 8 cm long, 6 cm wide and 5 cm high. Find AC.

Step 1: AC is the diagonal of the rectangular base with sides 8 and 6.

Step 2: $AC = \sqrt{8^2 + 6^2} = \sqrt{100} = 10 \text{ cm}$.

Final Answer: 10 cm.

12(b)(ii). Find AF, using the original cuboid labels (F is vertically above C).

Step 1: $AC = 10$ cm and $CF = 5$ cm are perpendicular. AF is the space diagonal.

Step 2: $AF = \sqrt{AC^2 + CF^2} = \sqrt{100 + 25} = \sqrt{125} = 5\sqrt{5}$ cm.

Final Answer: $5\sqrt{5}$ cm $\approx$ 11.18 cm.

12(b)(iii). Find angle CAF.

Step 1: In right triangle ACF , opposite side $CF = 5$ and adjacent side $AC = 10$.

Step 2: $\tan \angle CAF = 5/10 = 1/2$; the angle is $\arctan(1/2) \approx 26.565^\circ$.

Final Answer: $\angle CAF \approx 26.57^\circ$.

12(c). A ship sails from A(10°S,30°W) to B(11°N,30°W) at 900 km/h, departing 10:00 a.m. R = 6400 km. Find arrival time.

Step 1: Along the same meridian, angular distance $= 10 + 11 = 21^\circ$. Distance $= (21/360) \times 2\pi \times 6400 \approx 2345.72$ km.

Step 2: Travel time $= \text{distance}/900 \approx 2.60636$ hours $= 2$ hours 36.38 minutes.

Step 3: Add to 10:00 a.m. and round to the nearest minute: 12:36 p.m.

Final Answer: Approximately 12:36 p.m. the same day.

13(a). Use matrices to solve $2x - y = 5$ and $3x + 2y = 4$.

Step 1: Write $A\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ with $A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$. $\det A = 4 + 3 = 7$.

Step 2: $A^{-1} = (1/7)\begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$. Multiply by the right-hand vector: $(1/7)\begin{bmatrix} 14 \\ -7 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Step 3: Check: $2(2) - (-1) = 5$ and $3(2) + 2(-1) = 4$.

Final Answer: $x = 2$; $y = -1$.

13(b). Transform A(1,1), B(2,4), C(5,3) using $T = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$.

Step 1: Multiply each coordinate column by T : $(x', y') = (2x + y, x + 3y)$.

Step 2: $A' = (2 + 1, 1 + 3) = (3, 4)$; $B' = (4 + 4, 2 + 12) = (8, 14)$; $C' = (10 + 3, 5 + 9) = (13, 14)$.

Final Answer: $A'(3, 4)$, $B'(8, 14)$, $C'(13, 14)$.

13(c). Rotate A(4,2) by 120° anticlockwise about the origin.

Step 1: Rotation matrix is $[[\cos 120^\circ, -\sin 120^\circ], [\sin 120^\circ, \cos 120^\circ]] = [[-1/2, -\sqrt{3}/2], [\sqrt{3}/2, -1/2]]$.

Step 2: Multiply by (4,2): $x' = -2 - \sqrt{3}$; $y' = 2\sqrt{3} - 1$.

Final Answer: $A'(-2 - \sqrt{3}, 2\sqrt{3} - 1)$.

14(a)(i). Batiki uses 2.5 m, Kitenge 2 m; daily material limit 400 m. The paper says at most 80 metres of Batiki and at least 60 metres of Kitenge. Write constraints for garment counts x,y.

Step 1: Total material is $2.5x + 2y$; the capacity gives $2.5x + 2y \leq 400$.

Step 2: Taking the printed metre units literally, Batiki material $2.5x \leq 80$ gives $x \leq 32$; Kitenge material $2y \geq 60$ gives $y \geq 30$.

Step 3: The alternative interpretation often intended in this question uses 80 and 60 directly as numbers of garments. Keep the interpretations separate.

Final Answer: Literal metre wording: $x \geq 0, y \geq 0, 2.5x + 2y \leq 400, x \leq 32, y \geq 30$. **If 80 and 60 mean garment counts:** $x \leq 80, y \geq 60$ instead.

Study Tip: The wording mixes metres and garment counts. Both complete interpretations are provided rather than silently changing the units.

14(a)(ii). Maximise income $f(x,y) = 300x + 200y$ under those constraints.

Step 1: Literal feasible vertices: (0,30), (32,30), (32,160), (0,200). Their incomes are 6,000; 15,600; 41,600; 40,000 respectively.

Step 2: The largest is 41,600 at (32,160); material check $2.5(32) + 2(160) = 400$.

Step 3: Under the alternative garment-count limits, vertices are (0,60), (80,60), (80,100), (0,200), giving 12,000; 36,000; 44,000; 40,000. The maximum is 44,000 at (80,100).

Final Answer: Literal metre constraints: 32 Batiki and 160 Kitenge; maximum income 41,600. **If 80 and 60 are garment-count limits:** 80 Batiki and 100 Kitenge; maximum income 44,000.

14(b). Give two reasons for studying linear programming.

Step 1: Resource limits can be expressed as inequalities, such as time, labour or material limits.

Step 2: An objective function compares feasible choices, enabling a decision that meets the limits and optimises the desired result.

Final Answer: 1. It helps allocate limited resources efficiently. 2. It finds the greatest profit/output or least cost that satisfies stated constraints.

Section Summary: Grouped statistics are estimates based on midpoints and class boundaries. Match cuboid labels to the printed figure, preserve matrix order, and check linear-programming units before forming constraints.